

# Kosuke Ukita

PhD student (D1), full-time

Kyushu Institute of Technology

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## EDUCATION

Apr. 2026 - Present

### PhD

Kyushu Institute of Technology  
Department of Computer Science and Systems Engineering,  
Graduate School of Computer Science and Systems Engineering  
GPA: - / 4.0

Apr. 2024 - Mar. 2026

### MSc

Kyushu Institute of Technology  
Department of Creative Informatics,  
Graduate School of Computer Science and Systems Engineering  
Thesis: "A Study on Sensor Foundation Models via Joint Training  
using Flow Matching"  
GPA: 3.6 / 4.0

Apr. 2020 - Mar. 2024

### BSc

Kyushu Institute of Technology  
Department of Artificial Intelligence,  
Computer Science and Systems Engineering  
Thesis: "Proposal of Representation-Conditional Latent Diffusion  
Models: Hematoma Classification in Brain CT Images"  
GPA: 3.7 / 4.0

Apr. 2017 - Mar. 2020

### High School Diploma

Kanonji Daiichi High School (Designated as a Super Science High School)  
Science and Mathematics Course  
GPA: 4.9 / 5.0

## WORK EXPERIENCE

Dec. 2023 - Mar. 2026

### Research Assistant (RA)

Supervisor: Tsuyoshi Okita  
Researched and Developed foundation models

Apr. 2024 - Mar. 2027

### Teaching Assistant (TA)

Lecturer: Tsuyoshi Okita  
- Deep Learning  
- Deep Learning Exercise x2  
- Deep Learning Advanced I

Oct. 2023 - Nov. 2023

### OPTiM Corporation

## SOCIAL EXPERIENCE

- Nov. 2026 **Conference - Student Volunteer (SV)**  
The 14th International Symposium on Applied Engineering and Sciences (SAES 2026) · Fukuoka, Japan  
- Oral Session Chairperson  
- Other support
- Jun. 2026 **Conference - Student Volunteer (SV)**  
The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026) · Hong Kong, China  
- Registration desk  
- Session support  
- Excursion support  
- Other support

## PUBLICATIONS

- [1] **拡散モデルを用いた人間行動モーションデータからの写実的な映像生成**  
松石康希, 浮田嵩祐, 大北剛  
*情報処理学会論文誌 (IPSJ Journal)*, 67-10, 2026.  
July 2026 · Japan  
Refereed Journal
- [2] **Efficient Inference for Flow Matching via Unified Path CFG**  
Kosuke Ukita, Tsuyoshi Okita  
*The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)*  
June 2026 · Hong Kong, China  
Refereed Oral
- [3] **High-Performance Self-Supervised Learning by Joint Training of Flow Matching**  
Kosuke Ukita, Tsuyoshi Okita  
*The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)*  
May 2026 · Tangier, Morocco  
Refereed Spotlight (top 3%)
- [4] **From Sensor Data to Sequence of Scenes: Diffusion Model-Based Visualization of Human Activity**  
Koki Matsuishi, Kosuke Ukita, Tsuyoshi Okita

March 2026 · Hakodate, Japan

Refereed

Oral

[5] **Foundation Models to Tackle Activity Recognition in Unknown Domain: Sussex-Huawei Locomotion Challenge 2025 Task 2**

Tsuyoshi Okita, Kosuke Ukita, Asahi Miyazaki, Daichi Kubota, Jukichi Ota, Naoki Kagiya, Asahi Nishikawa, Daichi Nagayasu, Syunya Tomitaka, Daisuke Nozaki, Yuki Odo, Raku Yamashita, Xiaolong Ye, Huayu Gao, Kazuki Okahashi, Koki Matsuishi, Masaharu Kagiya, Kodai Hirata, Haruki Kai, Lin Wang, Hristijan Gjoreski, Mathias Ciliberto, Paula Lago, Kazuya Murao, Daniel Roggen

*Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '25)*, pp. 983–991

October 2025 · Espoo, Finland

Refereed

Oral

[6] **Multimodal Foundation Model for Cross-Modal Retrieval and Activity Recognition Tasks**

Koki Matsuishi, Kosuke Ukita, Tsuyoshi Okita

*International Journal of Activity and Behavior Computing (IJABC), Care XDX Center, Kyushu Institute of Technology, Vol. 2025, No. 3, pp. 1-25*

November 2025 · Kitakyushu, Japan

Refereed

Oral

[7] **Energy-Efficient High-Performance Representation Learning for Sensor Data via Flow Matching**

Kosuke Ukita, Tsuyoshi Okita

*The 7th International Symposium on Neuromorphic AI Hardware, Paper No. P-19*

October 2025 · Kitakyushu, Japan

Non-refereed

Poster

[8] **Image Classification Using a Diffusion Model as a PreTraining Model**

Kosuke Ukita, Xiaolong Ye, Tsuyoshi Okita

*arXiv preprint arXiv:2505.06890, pp. 1-10*

May 2025

Non-refereed

Preprint

- [9] **AURA-MFM: 詳細な行動理解のためのCLIPを用いたマルチモーダル基盤モデルの構築**

松石康希, 浮田嵩祐, 大北剛

情報処理学会研究報告(IPSJ SIG Technical Report), Vol. 2025-UBI-85, No. 8

February 2025 · Osaka, Japan

Non-refereed

Oral

- [10] **Improving Human Activity Recognition through Diffusion Models**

Kosuke Ukita, Tsuyoshi Okita

*The 12nd International Symposium on Applied Engineering and Sciences (SAES 2024)*, No. p0052

November 2024 · Fukuoka, Japan

Non-refereed

Oral

- [11] **Ensemble Learning of Models for Human Activity Recognition**

Asahi Miyazaki, Kosuke Ukita, Tsuyoshi Okita

*The 12nd International Symposium on Applied Engineering and Sciences (SAES 2024)*, No. p0053

November 2024 · Fukuoka, Japan

Non-refereed

Oral

- [12] **Analysis of Human Activity Recognition by Diffusion Models**

Kosuke Ukita, Tsuyoshi Okita

*Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '24)*, pp. 458–463

October 2024 · Melbourne, Australia

Refereed

Oral

- [13] **表現条件付き潜在拡散モデルと表現学習**

浮田嵩祐, Ye Xiaolong, 大北剛

情報処理学会研究報告, 火の国情報シンポジウム 2024, No. A4-4, pp. 1-8

March 2024 · Kagoshima, Japan

Non-refereed

Oral

- [14] **Towards LLMs for Sensor Data: Multi-Task Self-Supervised Learning**

Tsuyoshi Okita, Kosuke Ukita, Koki Matsuishi, Masaharu Kagiya, Kodai Hirata, Asahi Miyazaki

*Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct)*, pp. 499–504

October 2023 · Cancun, Mexico

Refereed

Oral

## GRANTS

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### – SPRING Scholarship Awardee

Japan Science and Technology Agency (JST)

April 2026 - March 2029 · JPY ¥1,800,000 (+ ¥900,000) / year

### – Student Travel Award

The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)

June 2026 · USD \$500

### – International Travel Grant

The Telecommunications Advancement Foundation

April 2026 · JPY ¥290,000

## AWARDS

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### – Student Travel Award

The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)

June 2026

### – Spotlight Presentation (Top 3%)

The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)

May 2026

### – Promising Researcher Award

Information Processing Society of Japan (IPSJ) Kyushu Branch

March 2024

### – Best Research Award

The Japan Statistical Society (JSS), Special Committee of Statistical Education (CSE),

The 8th Sports Data Analysis Competition

May 2019

## SKILLS

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### LANGUAGES

Python · C++ · C · Java · JavaScript · TypeScript · Go · SQL · R

### FRAMEWORKS & TOOLS

PyTorch · TensorFlow · Hugging Face · Jupyter · ABCI · Docker · Git

### OTHER SKILLS

Deep Learning · Neural Networks · Model Training · Data Processing · API Development · Linux